



Spring into AI

AI for Java Developers

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Introduction



About Me



Husband & Father



Cleveland, OH



Spring Developer Advocate



Java Champion

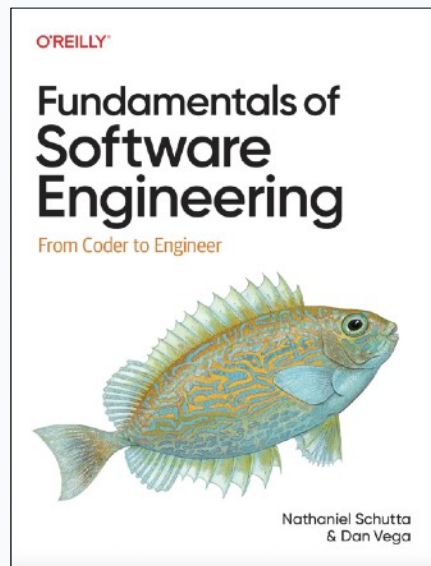


Author



24 Years in Software Development

danvega.dev





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The Java AI Opportunity



Why are we talking about AI on the JVM?



I know that most people won't believe it, but I can guarantee you that in 2 years, the majority of AI agent workloads will run on the JVM

James Ward

Principal Developer Advocate at AWS



Python earned its place in AI.

TensorFlow. PyTorch. Scikit-learn. Hugging Face.

Python dominated AI research and model training.
That part of the story is not up for debate.

But the game has changed.



But what are we actually doing?

Building & Training Models

- Specialized ML frameworks required
- Research-oriented workflows
- Customized Data Pipelines
- GPU Infrastructure

Integrating AI into Applications

- REST APIs & SDKs
- Authentication & Rate Limiting
- Retry Logic & Resilience
- Observability

Most of us aren't training models. **We're calling them.**

You've done this before.



LLMs are just the next Integration.

Java Is Built for This



Enterprise-grade infrastructure

Spring, Kafka, observability tools that are already running in production at scale



Type safety & structured output

Parse AI responses into real objects with compile-time guarantees, not loose strings



Battle-tested at scale

Millions of JVM applications serving billions of requests every single day



Your existing team

No need to hire a Python shop or replatform. Just upskill the developers you already trust

Calling OpenAI with cURL

```
#!/bin/bash
echo "Calling Open AI..."
MY_OPENAI_KEY="YOUR_API_KEY_HERE"
PROMPT="Tell me an interesting fact about Java"

curl https://api.openai.com/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $MY_OPENAI_KEY" \
  -d '{"model": "gpt-5", "messages": [{"role": "user", "content": "'${PROMPT}'"}] }'
```

API Response

```
{
  "id": "chatcmpl-ABNbjZ5oRbo720evnCX2arPufJCYK",
  "object": "chat.completion",
  "model": "gpt-5",
  "choices": [{
    "message": {
      "role": "assistant",
      "content": "Java was initially designed with interactive
        television in mind..."
    },
    "finish_reason": "stop"
  }],
  "usage": {
    "prompt_tokens": 14,
    "completion_tokens": 90,
    "total_tokens": 104
  }
}
```

Calling OpenAI with Java

```
public static void main(String[] args) throws IOException, InterruptedException {
    var apiKey = "YOUR_API_KEY_HERE";
    var body = """
        { "model": "gpt-5",
          "messages": [{
            "role": "user",
            "content": "Tell me an interesting fact about Java"
          }]
        }
    """;

    HttpRequest request = HttpRequest.newBuilder()
        .uri(URI.create("https://api.openai.com/v1/chat/completions"))
        .header("Content-Type", "application/json")
        .header("Authorization", "Bearer " + apiKey)
        .POST(HttpRequest.BodyPublishers.ofString(body))
        .build();

    var client = HttpClient.newHttpClient();
    var response = client.send(request, HttpResponse.BodyHandlers.ofString());
    System.out.println(response.body());
}
```

Spring AI provides us so much more
than a facility for making
REST API calls



What Are the Challenges?

What does a framework provide you?

- Model abstraction and portability
- Unified API across providers
- Structured output parsing
- Prompt template management
- Token counts & cost management
- Retry logic and error handling
- Response streaming and async processing
- Memory and conversation management
- Embedding and vector operations
- Function calling integration
- Observability and monitoring
- Security and compliance

LLM Pricing



ChatGPT Pricing Tiers

Free

Intelligence for everyday tasks

\$0 / month

[Get Free ↗](#)

- ✓ Limited access to flagship model GPT-5.2
- ✓ Limited messages and uploads
- ✓ Limited and slower image generation
- ✓ Limited deep research
- ✓ Limited memory and context

Have an existing plan? See [billing help](#)

Go

Keep chatting with expanded access

\$8 / month

[Get Go ↗](#)

◇ Everything in Free and:

- ✓ More access to our flagship model GPT-5.2
- ✓ More messages
- ✓ More uploads
- ✓ More image creation
- ✓ Longer memory

This plan may include ads. [Learn more](#)

Plus

Do more with advanced intelligence

\$20 / month

[Get Plus ↗](#)

◇ Everything in Go and:

- ✓ Advanced reasoning models
- ✓ Expanded messages and uploads
- ✓ Expanded and faster image creation
- ✓ Expanded deep research and agent mode
- ✓ Expanded memory and context
- ✓ Projects, tasks, and custom GPTs
- ✓ Codex agent and Sora video generation
- ✓ Early access to new features

Pro

Full access to the best of ChatGPT

\$200 / month

[Get Pro ↗](#)

◇ Everything in Plus and:

- ✓ Pro reasoning with GPT-5.2 Pro
- ✓ Unlimited GPT-5.2 and file uploads
- ✓ Unlimited and faster image creation
- ✓ Maximum deep research and agent mode
- ✓ Maximum memory and context
- ✓ Expanded projects, tasks, and custom GPTs
- ✓ Expanded access to Sora video generation
- ✓ Expanded, priority-speed Codex agent
- ✓ Research preview of new features

Unlimited subject to abuse guardrails. [Learn more](#)

API Pricing

GPT-5.2

The best model for coding and agentic tasks across industries

Price

Input:

\$1.750 / 1M tokens

Cached input:

\$0.175 / 1M tokens

Output:

\$14.000 / 1M tokens

GPT-5.2 pro

The smartest and most precise model

Price

Input:

\$21.00 / 1M tokens

Cached input:

-

Output:

\$168.00 / 1M tokens

GPT-5 mini

A faster, cheaper version of GPT-5 for well-defined tasks

Price

Input:

\$0.250 / 1M tokens

Cached input:

\$0.025 / 1M tokens

Output:

\$2.000 / 1M tokens

What is a Token?

~100 tokens

≈ 75 words

A token is roughly 3/4 of a word

What is a Token?

GPT-5.x & O1/3

GPT-4 & GPT-3.5 (legacy)

GPT-3 (legacy)

Java is still one of the most widely deployed languages in the world, running billions of devices—from ATMs to enterprise backends to parts of the Android ecosystem.

Clear

Show example

Tokens

32

Characters

165

Java is still one of the most widely deployed languages in the world,
running billions of devices—from ATMs to enterprise backends to parts of
the Android ecosystem.

Text

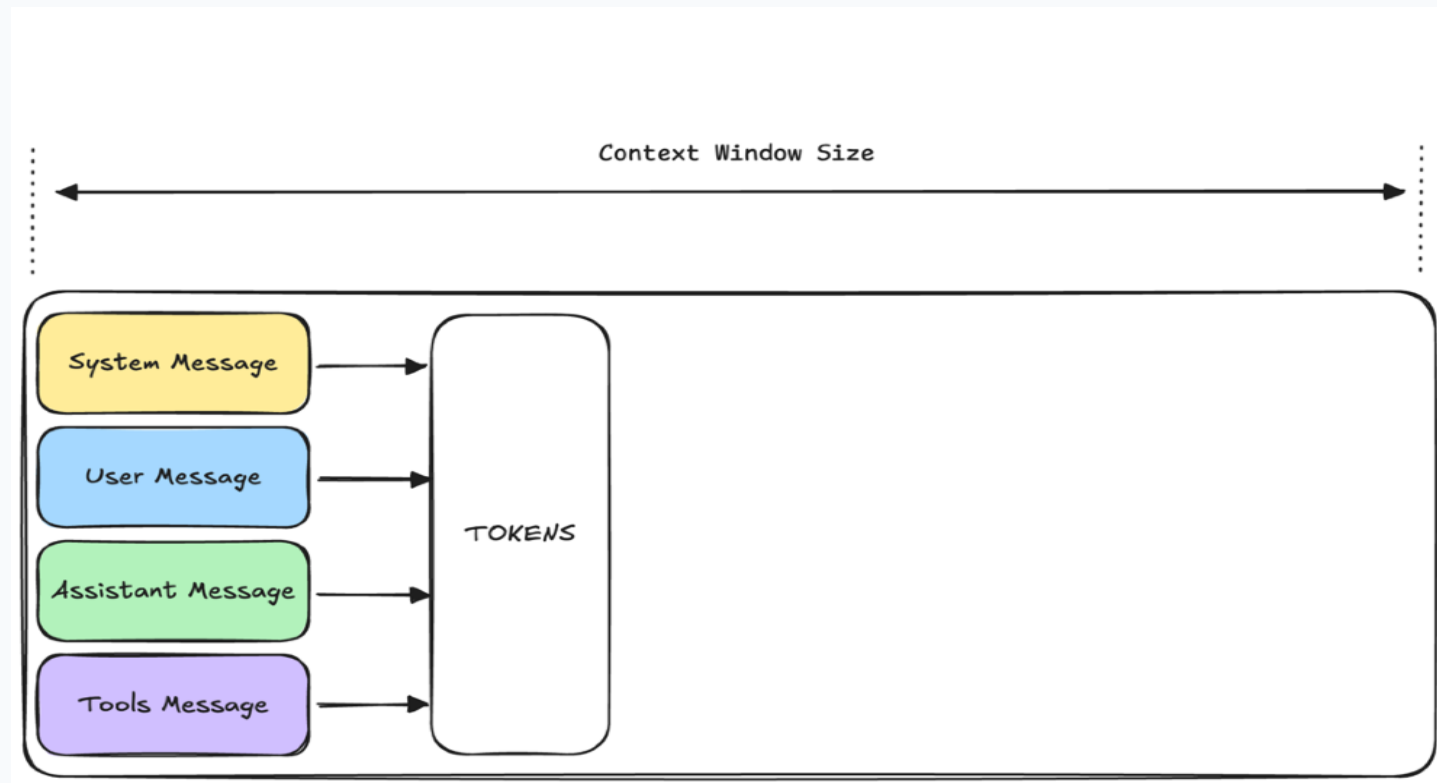
Token IDs

[21220, 382, 2928, 1001, 328, 290, 1645, 20360, 51933, 22772, 306, 290,
2375, 11, 6788, 68167, 328, 10220, 178328, 11503, 25153, 316, 22459, 1602,
1847, 316, 7881, 328, 290, 9438, 38423, 13]

Text

Token IDs

Context Window



LLM Pricing Comparison

Model	Context	Input / Output (per 1M tokens)
GPT-5 (OpenAI)	~400K	\$1.25 / \$10.00
GPT-5 Mini	~400K	\$0.25 / \$2.00
GPT-5 Nano	~400K	\$0.05 / \$0.40
Claude Sonnet 4	200K	\$3.00 / \$15.00
Claude Opus 4.1	200K (32K out)	\$15.00 / \$75.00
Gemini 2.5 Flash-Lite	1M	\$0.10 / \$0.40
Gemini 2.5 Flash	1M	\$0.30 / \$1.25
Gemini 2.5 Pro	1M (2M roadmap)	\$1.25-2.50 / \$10-15
Grok 3 (xAI)	131K	\$3.00 / \$15.00

Spring AI Introduction



Getting Started

A Tour of Spring AI Features

- Spring AI Reference Documentation
- Currently Versions:
 - Spring Boot 3 - v1.1.2
 - Spring Boot 4 - 2.0.0 M2
- start.spring.io
- Chat Client & Chat Model
- Blocking vs Non-Blocking (Streaming Responses)
- Response Types (Content / ChatResponse)
- Spring AI Features (Prompt Templates, Structured Output & more)



Demo Time

Check out the code



Thank You

www.danvega.dev